Coh Category

Verifier-Certified Trace Composition and Projection-Induced Semantic Cost

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Abstract

We define the Coh category, a categorical structure in which morphisms are not assumed as primitive arrows but are generated by verifier-certified traces. A trace is admissible only when it satisfies a resource inequality of the form

$$V(y) + \mathrm{Spend}(\tau) \leq V(x) + \mathrm{Defect}(\tau).$$

We then add a projection layer: hidden realizations project to observable traces, and a semantic defect is induced by taking the worst-case hidden cost over the corresponding fiber. This projection-induced envelope turns additive hidden cost into subadditive observable cost. The resulting certified traces form a category whose hom-sets carry a natural preorder by defect.

1 Purpose and informal reading

The guiding idea is simple: in a verification-sensitive system, a transition should not count as an available arrow merely because two states exist. It should count as a morphism only after it has passed a deterministic admissibility rule.

Thus Coh reverses the usual order of construction:

not arrow first, verification later,

but verification first, morphism second.

The categorical claim is that this verifier-generated collection of admissible traces is still closed under identity and composition, provided the verifier and bookkeeping laws satisfy the stated hypotheses.

2 Operational layer: certified traces

Definition 2.1 (Coh operational system)**.** *A Coh operational system consists of:*

- *a state space X;*
- *a hidden state space H;*
- *a projection $\Pi : H \to X$;*
- *a deterministic verifier RV;*
- *a valuation $V : X \to \mathbb{Q}_{\geq 0}$;*

- *finite traces* $\tau : x \rightsquigarrow y$ *between states;*
- *exact nonnegative rational bookkeeping quantities* $\mathrm{Spend}(\tau)$ *and* $\mathrm{Defect}(\tau)$.

Definition 2.2 (Admissible trace)**.** *A trace* $\tau : x \rightsquigarrow y$ *is* admissible *when:*

1. *it is accepted by the deterministic verifier* RV*;*
2. *it satisfies the resource inequality*

$$V(y) + \mathrm{Spend}(\tau) \leq V(x) + \mathrm{Defect}(\tau).$$

Remark 2.3 (Spend and defect)**.** *The two channels have different meanings.* Spend *records cost paid by the transition.* Defect *records allowed slack, tolerance, or uncertainty. Collapsing them into one number would remove the distinction between controlled cost and tolerated error.*

3 The category of certified traces

Definition 3.1 (Coh hom-set)**.** *For states* $x, y \in X$*, define*

$$\mathrm{Hom}_{\mathbf{Coh}}(x, y) := \{\tau : x \rightsquigarrow y \mid \tau \text{ is admissible}\}.$$

The identity morphism at x *is the empty trace* $\epsilon_x : x \rightsquigarrow x$*. If* $\tau_1 : x \rightsquigarrow y$ *and* $\tau_2 : y \rightsquigarrow z$ *are admissible, their composition is trace concatenation:*

$$\tau_2 \circ \tau_1 := \tau_2 * \tau_1.$$

Axiom 3.2 (Verifier identity and closure)**.** *The verifier accepts empty traces, and accepted traces are closed under compatible concatenation. That is, if* $RV(\tau_1) = \mathrm{ACCEPT}$ *and* $RV(\tau_2) = \mathrm{ACCEPT}$ *for composable traces, then*

$$RV(\tau_2 * \tau_1) = \mathrm{ACCEPT}.$$

Axiom 3.3 (Additive bookkeeping)**.** *For composable traces,*

$$\mathrm{Spend}(\tau_2 * \tau_1) = \mathrm{Spend}(\tau_1) + \mathrm{Spend}(\tau_2),$$

$$\mathrm{Defect}(\tau_2 * \tau_1) = \mathrm{Defect}(\tau_1) + \mathrm{Defect}(\tau_2).$$

Theorem 3.4 (Certified traces form a category)**.** *Assume verifier identity, verifier closure, and additive bookkeeping. Then admissible traces form a category* **Coh**.

Proof. The identity at x is the empty trace ϵ_x. Its spend and defect are both zero, so the governing inequality reads

$$V(x) + 0 \leq V(x) + 0,$$

which holds. By verifier identity, ϵ_x is accepted, hence admissible.

Now let $\tau_1 : x \rightsquigarrow y$ and $\tau_2 : y \rightsquigarrow z$ be admissible. Their certified inequalities are

$$V(y) + \mathrm{Spend}(\tau_1) \leq V(x) + \mathrm{Defect}(\tau_1),$$

$$V(z) + \mathrm{Spend}(\tau_2) \leq V(y) + \mathrm{Defect}(\tau_2).$$

Adding these inequalities and canceling the finite term $V(y)$ gives

$$V(z) + \mathrm{Spend}(\tau_1) + \mathrm{Spend}(\tau_2) \leq V(x) + \mathrm{Defect}(\tau_1) + \mathrm{Defect}(\tau_2).$$

By additive bookkeeping, this is precisely

$$V(z) + \mathrm{Spend}(\tau_2 * \tau_1) \leq V(x) + \mathrm{Defect}(\tau_2 * \tau_1).$$

Verifier closure gives acceptance of $\tau_2 * \tau_1$, so the composite is admissible. Associativity follows from associativity of finite trace concatenation, and the empty trace gives the unit laws. □

$$\begin{tikzcd} x \arrow[r, "\tau_1"] \arrow[rr, bend left, "\tau_2 \circ \tau_1 = \tau_2 * \tau_1"] & y \arrow[r, "\tau_2"] & z \end{tikzcd}$$

Figure 1: Certified composition is trace concatenation.

4 Projection and semantic cost

The operational category above uses exact syntactic bookkeeping. Coh also distinguishes this from semantic cost induced by hidden realizations.

Definition 4.1 (Hidden realizations of an observable trace)**.** *For an observable trace τ, let $R(\tau)$ denote the finite set of verifier-realizable hidden traces projecting to τ.*

Definition 4.2 (Semantic cost)**.** *Let W be a hidden trajectory cost. Define*

$$s_{\mathrm{sem}}(\tau) := \max_{\Theta \in R(\tau)} W(\Theta).$$

Equivalently, in the abstract V2 envelope notation, if $\mathcal{F}_\tau$ is the fiber over τ, then

$$\delta(\tau) = \sup_{\Theta \in \mathcal{F}_\tau} W(\Theta).$$

For finite fibers, the supremum is a maximum, so $\delta(\tau) = s_{\mathrm{sem}}(\tau)$.

Axiom 4.3 (Hidden cost subadditivity)**.** *For composable hidden realizations,*

$$W(\Theta_2 * \Theta_1) \leq W(\Theta_1) + W(\Theta_2).$$

Axiom 4.4 (Restriction of composite realizations)**.** *Every hidden realization of a composite observable trace restricts to hidden realizations of its component traces. Thus if*

$$\Theta \in R(\tau_2 * \tau_1),$$

then there exist

$$\Theta_1 \in R(\tau_1), \qquad \Theta_2 \in R(\tau_2)$$

compatible with Θ.

Theorem 4.5 (Projection-induced subadditivity)**.** *For composable admissible traces,*

$$s_{\mathrm{sem}}(\tau_2 \circ \tau_1) \leq s_{\mathrm{sem}}(\tau_1) + s_{\mathrm{sem}}(\tau_2).$$

Equivalently, in V2 notation,

$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2).$$

Proof. Let $\Theta \in R(\tau_2 * \tau_1)$. By the restriction axiom, Θ induces component realizations

$$\Theta_1 \in R(\tau_1), \qquad \Theta_2 \in R(\tau_2).$$

By hidden cost subadditivity,

$$W(\Theta) \leq W(\Theta_1) + W(\Theta_2).$$

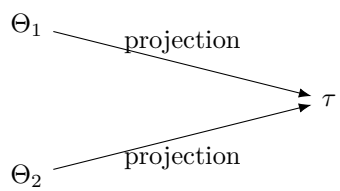
Since $s_{\mathrm{sem}}(\tau_i)$ is the maximum hidden cost over $R(\tau_i)$, we have

$$W(\Theta_1) \leq s_{\mathrm{sem}}(\tau_1), \qquad W(\Theta_2) \leq s_{\mathrm{sem}}(\tau_2).$$

Therefore

$$W(\Theta) \leq s_{\mathrm{sem}}(\tau_1) + s_{\mathrm{sem}}(\tau_2).$$

Taking the maximum over all $\Theta \in R(\tau_2 * \tau_1)$ yields the result. □



multiple hidden realizations can collapse to one observable trace

Figure 2: Projection collapses hidden realizations into observable traces.

5 Worked example: strict separation

The framework is useful because syntactic bookkeeping and semantic cost need not coincide.

Proposition 5.1 (Strict separation)**.** *There exist admissible traces τ such that*

$$s_{\mathrm{sem}}(\tau) < s_{\mathrm{syn}}(\tau).$$

Proof. Consider a trace $\tau = \sigma_2 * \sigma_1$ with syntactic step spends

$$s(\sigma_1) = 1, \qquad s(\sigma_2) = 1.$$

Then

$$s_{\mathrm{syn}}(\tau) = s(\sigma_1) + s(\sigma_2) = 2.$$

Suppose the composite observable trace has a single verifier-realizable hidden realization Θ_* with

$$W(\Theta_*) = 1.$$

Then

$$s_{\mathrm{sem}}(\tau) = \max_{\Theta \in R(\tau)} W(\Theta) = W(\Theta_*) = 1.$$

Hence

$$s_{\mathrm{sem}}(\tau) = 1 < 2 = s_{\mathrm{syn}}(\tau).$$

□

This example represents hidden reuse, cancellation, or compression that is not visible in the syntactic ledger. The syntactic ledger remains safe and compositional, while the semantic layer records a tighter realized cost.

6 V1-to-V2 dictionary

The abstract V2 terminology is obtained from the operational V1 system by the following identifications:

V1 operational term	V2 projection term
observable trace τ	observable trace R
$R(\tau)$ hidden realizations	$\mathcal{F}_R$ hidden fiber
$W(\Theta)$	hidden cost
$s_{\mathrm{sem}}(\tau)$	$\delta(R)$
admissible trace	certified morphism

Thus the V2 envelope is not a separate idea: it is the abstract form of the V1 semantic cost construction.

7 Conclusion

The Coh category formalizes verifier-generated morphisms: traces become arrows only after satisfying deterministic admissibility and a resource inequality. The projection layer then explains why observable semantic cost is subadditive. Hidden costs may compose additively or subadditively, while projection to observables forces the certified layer to reason through worst-case fibers.

The main structural conclusion is:

$$\boxed{\text{local verified constraints compose into global categorical structure.}}$$

References

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